LINUX RECYCLE BIN PROJECT SUBMISSION COVER SHEET

STUDENTS INFORMATION

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Course: Engenharia Informática
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PROJECT INFORMATION

Project Title: Linux Recycle Bin System
Version: 75 (final commit)
Total files: 131

- recycle_bin.sh
- test_suite.sh + 1 auxiliriary script
- README.md
- TESTING.md
- TECHNICAL_DOC.md
- screenshots: 106
- diagrams: 19

Archive Size: 5.9MB

IMPLEMENTATION STATUS

Core Features:
✔ Initialize recycle bin
✔ Delete files/directories
✔ List recycle bin contents
✔ Restore files
✔ Search functionality
✔ Empty recycle bin
✔ Help system

Optional Features Implemented:
✔ Statistics dashboard
✔ Auto-cleanup
✔ Quota management
✔ File preview
✔ Other: Lock operation, Secure operation, Check metadata corruption, Repair metadata file.

TESTING STATUS

Total Test Cases: 44 (manual) / 53 (test suite)
Tests Passed: 44 (manual) / 53 (test suite)
Tests Failed: 0
Pass Rate: 100%

KNOWN ISSUES

We are unaware of any limitation or bug with our system.

COLLABORATION DECLARATION

We declare that this work is our own and has been completed in accordance with the academic integrity policy. We have properly cited all sources and assistance received.

Resources Used:

- silence mv: https://unix.stackexchange.com/questions/87605/is-there-a-way-to-make-mv-fail-silently
- read line by line: https://stackoverflow.com/questions/10929453/read-a-file-line-by-line-assigning-the-value-to-a-variable
- readable file size: https://stackoverflow.com/questions/15854332/file-size-in-human-readable-format
- remove lines: https://stackoverflow.com/questions/5410757/how-to-delete-from-a-text-file-all-lines-that-contain-a-specific-string
- free stdin: https://unix.stackexchange.com/questions/460266/use-read-as-a-prompt-inside-a-while-loop-driven-by-read
- grep to search patterns: https://www.w3schools.com/bash/bash_grep.php
- regex with sed: https://stackoverflow.com/questions/29613304/is-it-possible-to-escape-regex-metacharacters-reliably-with-sed
- config file source idea: https://askubuntu.com/questions/743493/best-way-to-read-a-config-file-in-bash
- date epoch: https://schuchert.github.io/daily%20tech/2021/02/14/SedBreakdown.html?utm
- read -s: https://www.geeksforgeeks.org/linux-unix/bash-script-read-user-input/
- regex explantion (for secure_operation()): https://stackoverflow.com/questions/48594826/bash-conditional-loop-issue-y-or-y-yes-n-or-n-no

- locking shell script: https://unix.stackexchange.com/questions/22044/correct-locking-in-shell-scripts

AI Assistance:
It's important to note that LLMs were used as a support tool to comprehend concepts, to study functionalities of commands identified during research and any debugging that occurred during the development, as well as helping refine and translate concepts to english.

STUDENTS SIGNATURE





Date: 31/10/2025